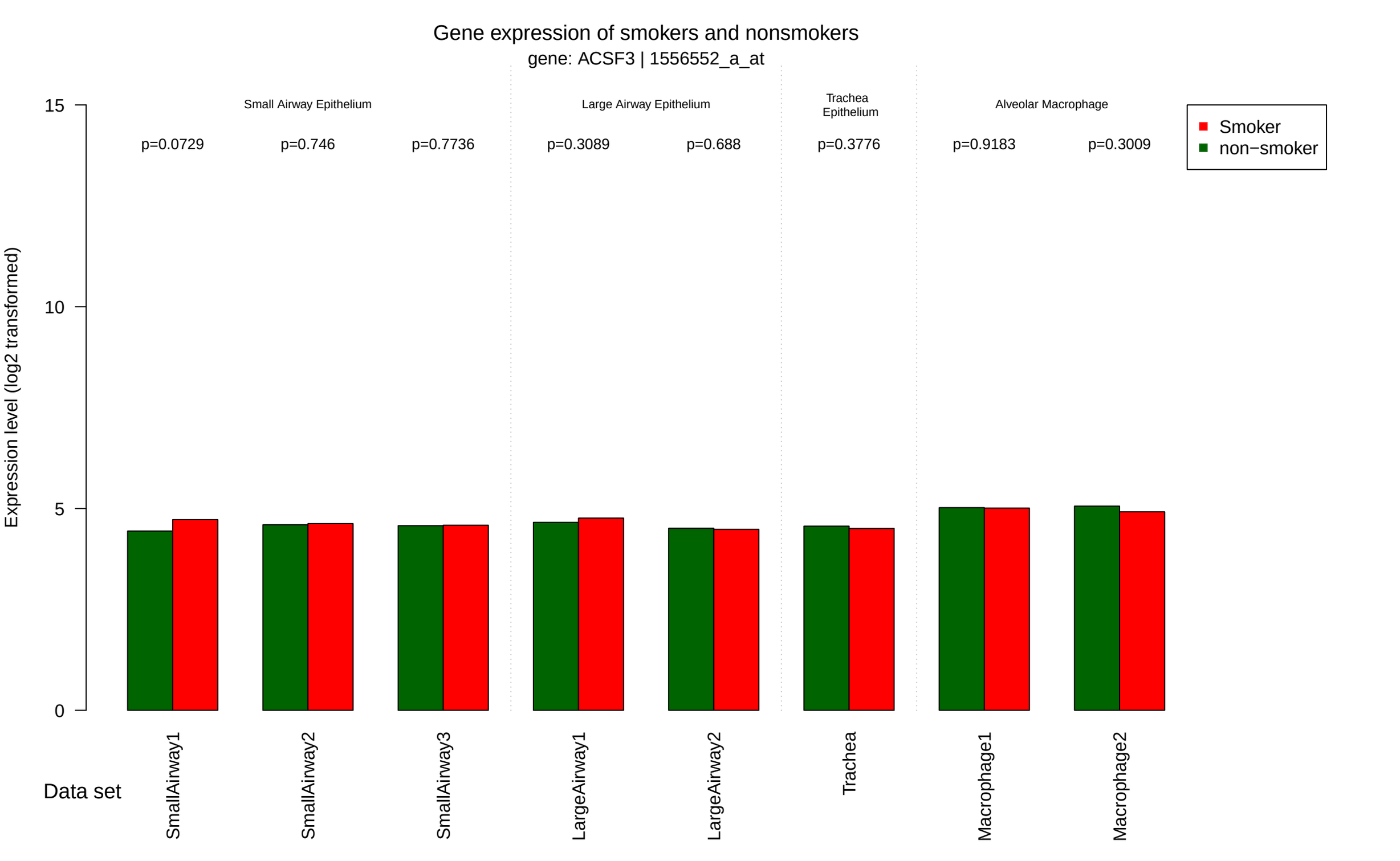
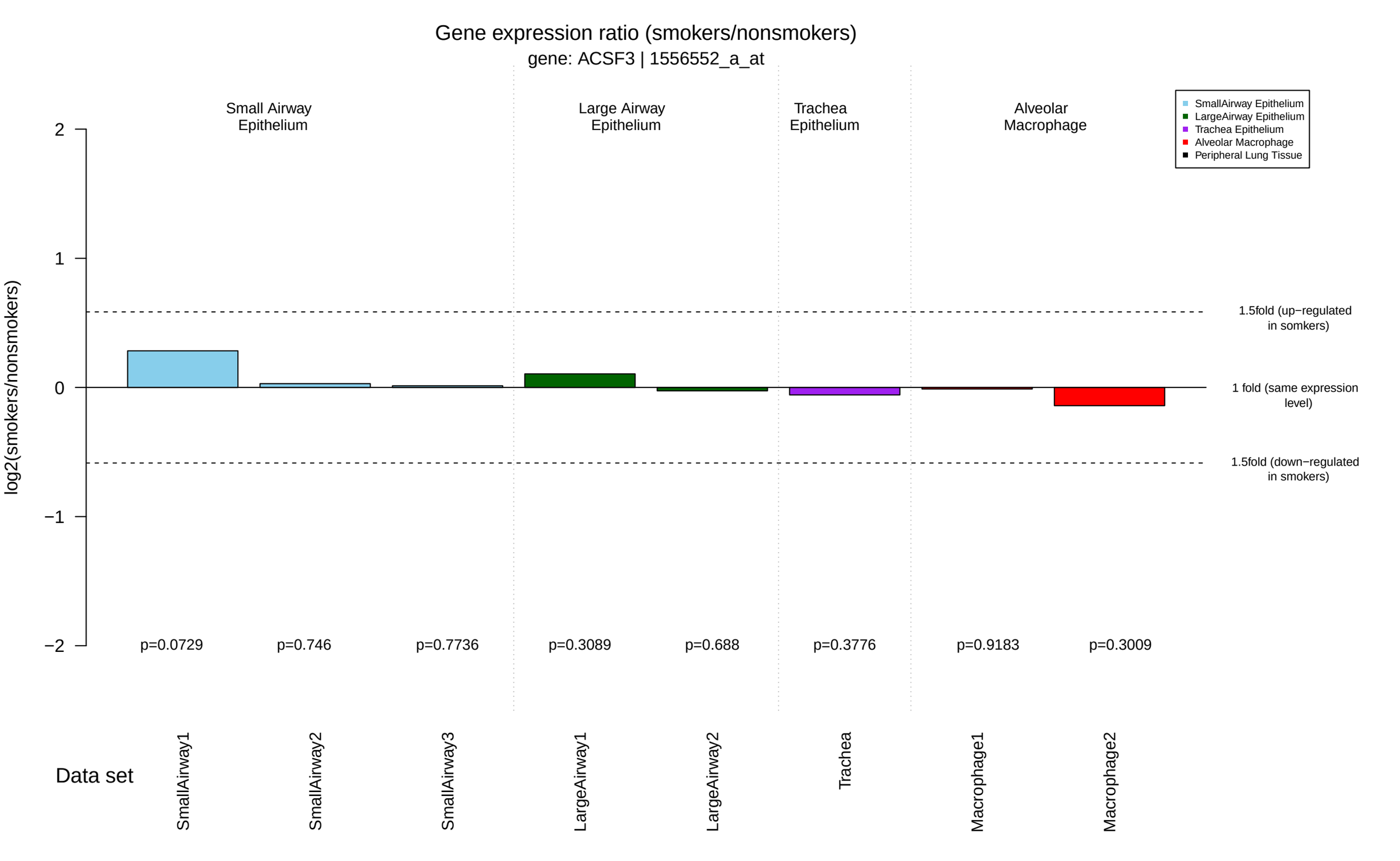
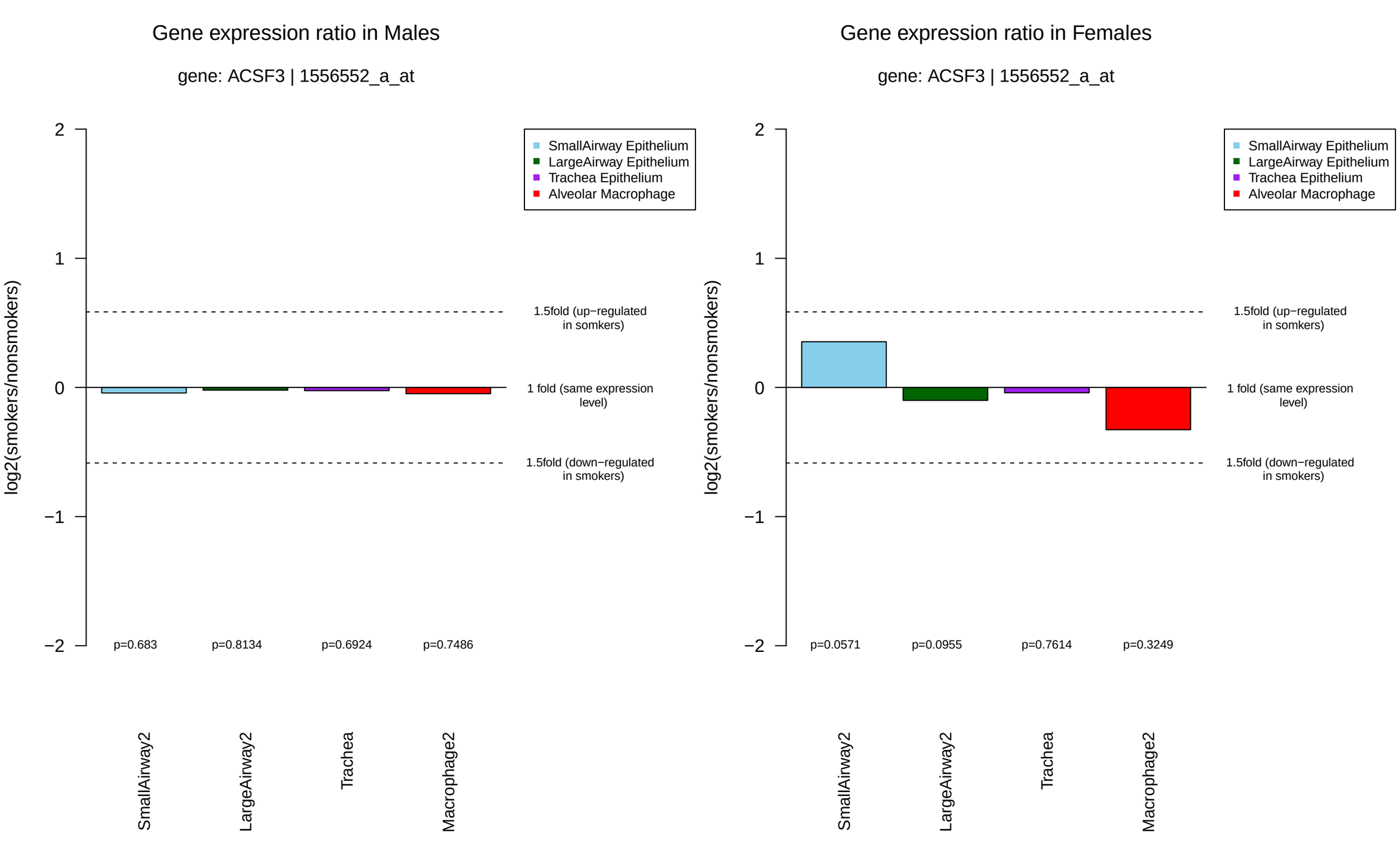
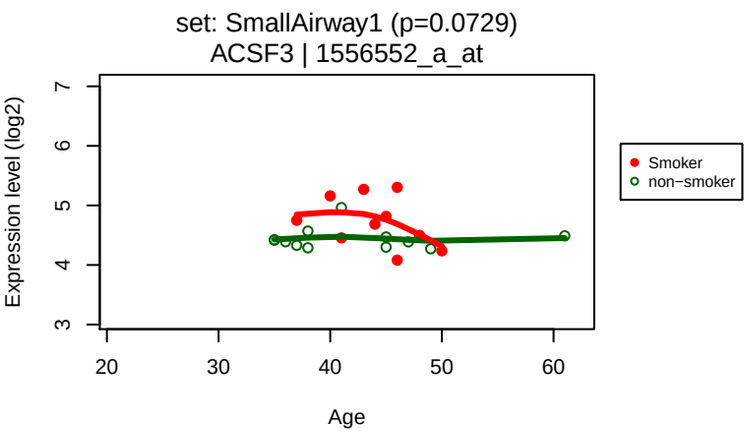
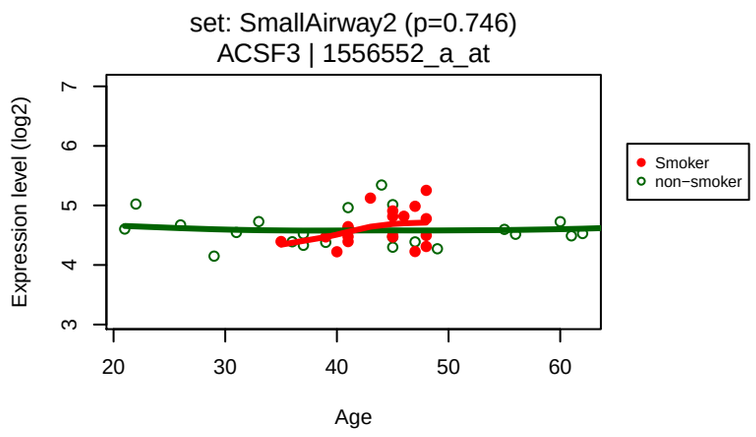
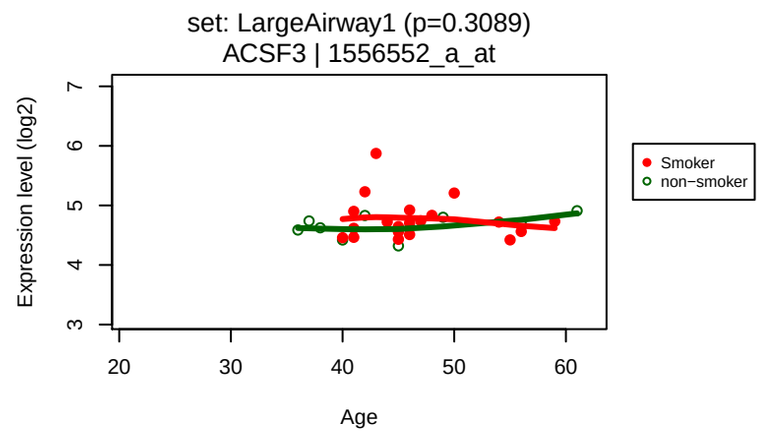

Gene expression of smokers and nonsmokers
gene: ACSF3 | 1556552_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0729
p=0.746
p=0.7736
p=0.3089
p=0.688
p=0.3776
p=0.9183
p=0.3009
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

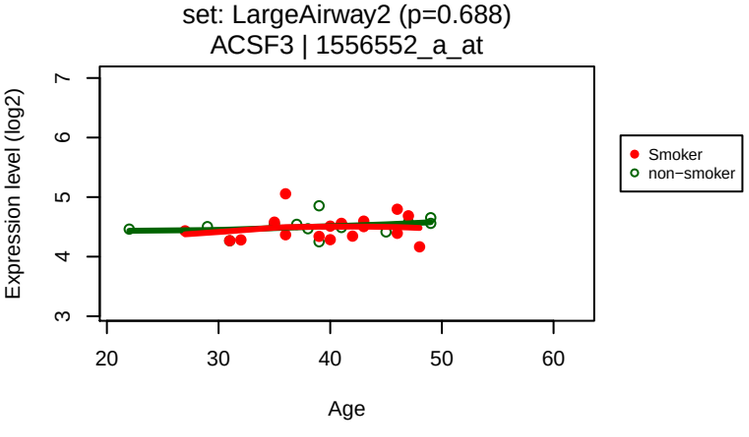

Gene expression ratio (smokers/nonsmokers)
gene: ACSF3 | 1556552_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0729
p=0.746
p=0.7736
p=0.3089
p=0.688
p=0.3776
p=0.9183
p=0.3009
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

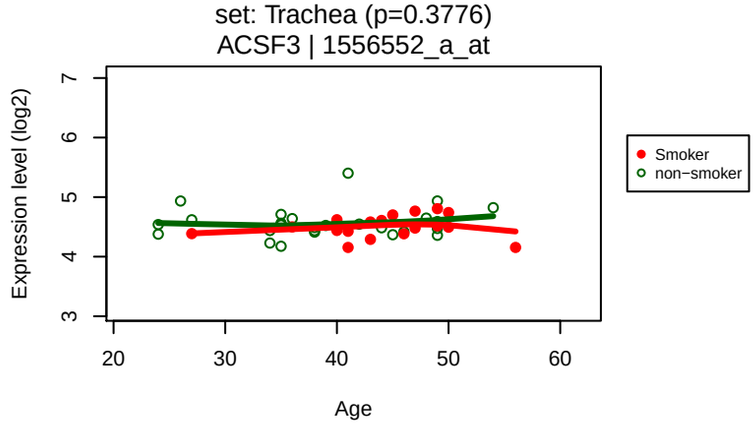

Gene expression ratio in Males
gene: ACSF3 | 1556552_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.683
p=0.8134
p=0.6924
p=0.7486
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ACSF3 | 1556552_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0571
p=0.0955
p=0.7614
p=0.3249
SmallAirway2
LargeAirway2
Trachea
Macrophage2

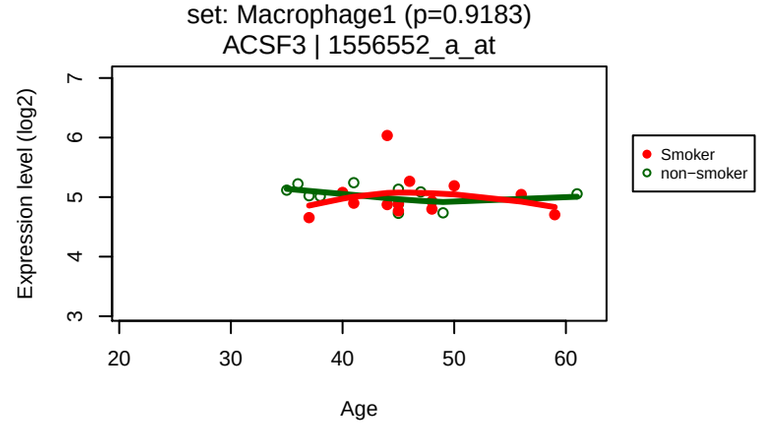

set: SmallAirway1 (p=0.0729)
ACSF3 | 1556552_a_at
Expression level (log2)
Age
Smoker
non−smoker

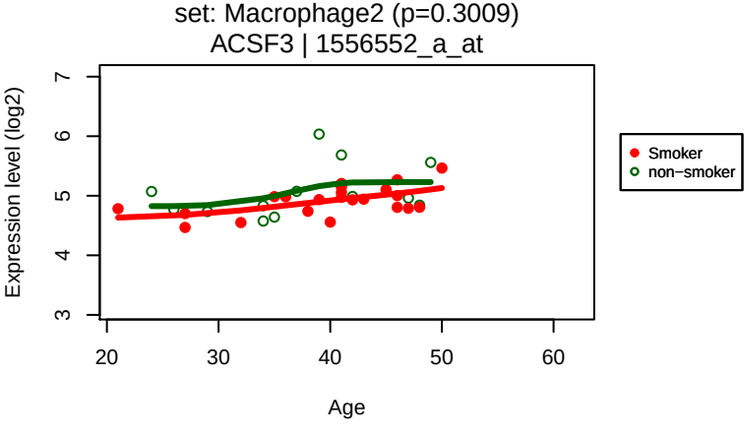

set: SmallAirway2 (p=0.746)
ACSF3 | 1556552_a_at
Expression level (log2)
Age
Smoker
non−smoker

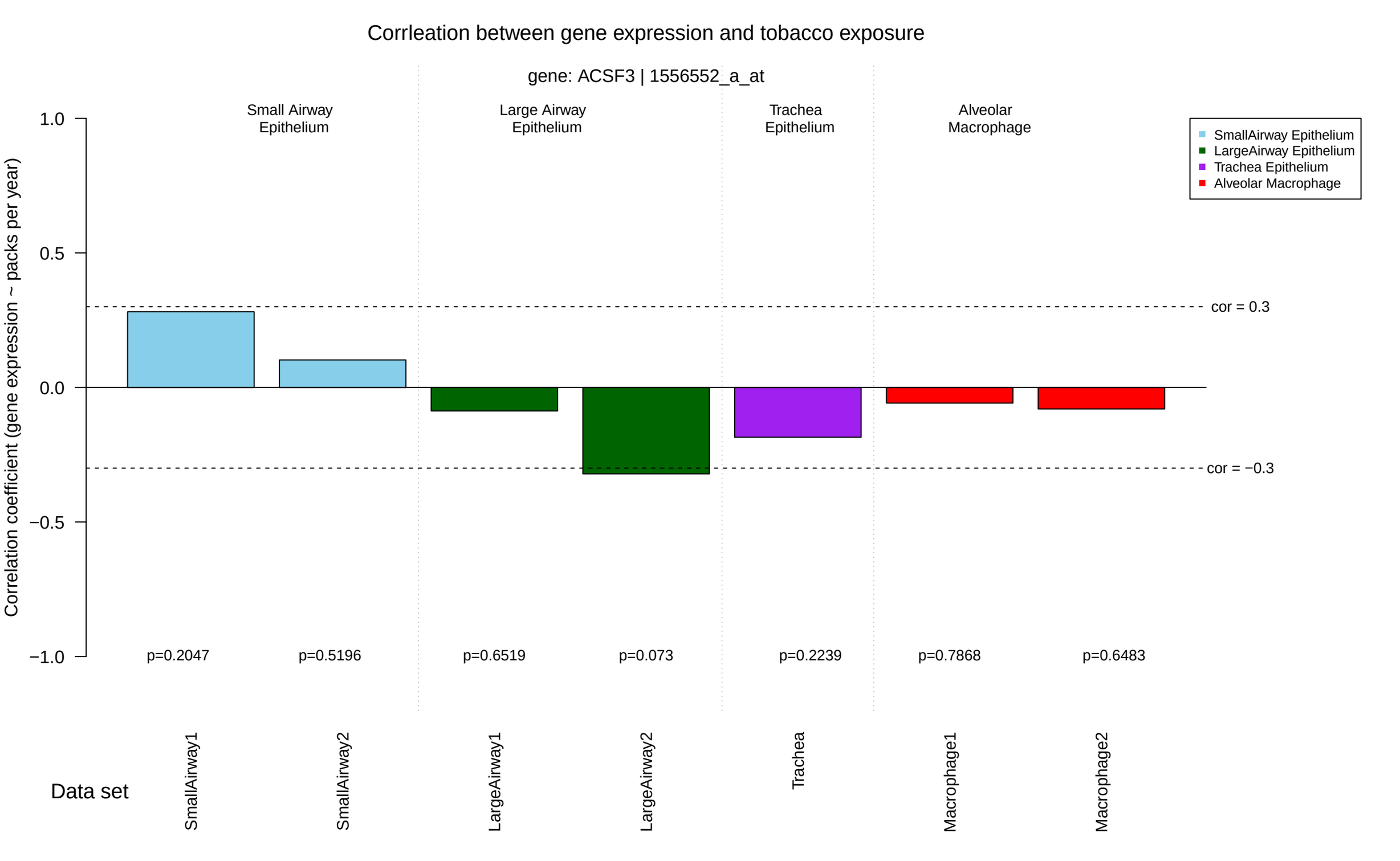

set: LargeAirway1 (p=0.3089)
ACSF3 | 1556552_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.688)
ACSF3 | 1556552_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3776)
ACSF3 | 1556552_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9183)
ACSF3 | 1556552_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3009)
ACSF3 | 1556552_a_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ACSF3 | 1556552_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2047
p=0.5196
p=0.6519
p=0.073
p=0.2239
p=0.7868
p=0.6483
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2